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Modular welding positioner

Abstract

A modular welding positioner is disclosed. The modular welding positioner includes a base, the base including: spaced-apart guides positioned thereon, the guides being arranged so as to define both a circular path and a linear path; and a driving element to move a positioner positioned on the base.

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Background/Summary

BACKGROUND OF THE INVENTION

(1) This invention relates generally to a welding apparatus, and more particularly to a modular welding positioner for use in a welding process.

(2) Welding positioners are well-known in the welding industry. Welding positioners allow a user to secure a device to the welding positioner and perform a weld. For example, a lot of the welding

positioners are rotary positioners to allow a user to clamp a workpiece thereto and perform a circumferential weld. By having a uniform rotation of the workpiece, the user is able to provide a uniform and strong weld around the circumference of the workpiece. Another type of welding positioner is a linear positioner.

(3) Unfortunately, not all welding is performed either circumferentially or linearly and as a result, users are required to buy multiple welding positioners to perform a single welding job.

(4) Accordingly, there is a need for a welding positioner that allows a user to perform both linear and circumferential welds using a single welding positioner.

BRIEF SUMMARY OF THE INVENTION

(5) This need is addressed by the present invention, which provides a modular welding positioner that allows a user to perform both linear and circumferential welds.

(6) According to an aspect of the invention, a modular welding positioner includes a base, the base including: spaced-apart guides positioned thereon, the guides being arranged so as to define both a circular path and a linear path; and a driving element to move a positioner positioned on the base.

(7) According to another aspect of the invention, a modular welding positioner includes a rotary positioner configured to rotate an object to be welded; a linear positioner configured to move linearly an object to be welded; and a base having a plurality of guides positioned thereon for receiving one of the rotary positioner or linear positioner; wherein the modular welding positioner is configurable between a first configuration where the rotary positioner is positioned between the guides on the base and a second configuration where the linear positioner is positioned between the guides on the base.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention may be best understood by reference to the following description taken in conjunction with the accompanying drawing figures, in which:

(2) FIG. 1 shows a modular welding positioner according to an embodiment of the invention;

(3) FIG. 2 shows a base of the modular welding positioner of FIG. 1;

(4) FIG. 3 shows a bottom view of the base of FIG. 2;

(5) FIG. 4 shows a top view of the base of FIG. 2;

(6) FIG. 5 shows a first configuration of the welding positioner of FIG. 1;

(7) FIG. 6 shows a side elevation of the welding positioner of FIG. 5;

(8) FIG. 7 shows a top view of the welding positioner of FIG. 5 with jaws in a first position;

(9) FIG. 8 shows a top view of the welding positioner of FIG. 5 with jaws in a second position;

(10) FIG. 9 shows a second configuration of the welding positioner of FIG. 1;

(11) FIG. 10 shows a perspective view of the welding positioner of FIG. 9; and

(12) FIG. 11 shows a side elevation of the welding positioner of FIG. 9.

DETAILED DESCRIPTION OF THE INVENTION

(13) Referring to the drawings wherein identical reference numerals denote the same elements throughout the various views, FIG. 1 illustrates an exemplary modular welding positioner 10. The modular welding positioner 10 includes a base 12, a rotary positioner 14, and a linear positioner 16.

(14) As shown in FIGS. 2-4, the base 12 includes an upper platform 18 supported by a sidewall 20. The platform 18 is in the form of a ring, i.e., the platform 18 includes an aperture 22 extending therethrough. Guides 24 are spaced along a periphery of the platform 18 for receiving the rotary positioner 14 or linear positioner 16 therebetween. As shown, four guides 24 are equally spaced along a periphery of the platform 18; however, it should be appreciated that any suitable combination of guides 24 and spacing may be used. Further, it should be appreciated that the guides 24 may act as stationary guides (i.e., do not rotate) or as rollers (i.e., rotate). The guides 24

may be made of any suitable material such as plastic (for example, polytetrafluoroethylene (PTFE)), metal (for example, aluminum and steel) and a friction material such as rubber.

(15) The base further includes a gate **26** pivotally connected to the sidewall **20** at hinge **32**. Any suitable hinge or pivot may be used to allow the gate **26** to pivot relative to the sidewall **20**. The gate **26** supports a motor **28** used to operate a driving element such as gear **30**. It should be appreciated that a friction element may also be considered a gear. One example of a motor is a step motor; however, it should be appreciated that any suitable motor may be used to operate gear **30**. The gate **26** pivots between a use position where the gear **30** is engaged with a gear (discussed below) of the rotary positioner **14** or of the linear positioner **16** and a non-use position where the gear **30** is no longer engaged. The platform **18** includes a cutout or slot **38** to allow a shaft **34** of the gear **30** to move inwardly toward a center of the platform **18** when the gate **26** is moved to the use position. A spring **36** biases the gate **26** in the use position. Base mounts **40** extend from the sidewall to allow the base to be secured to a table or other device using fasteners such as bolts.

(16) Referring to FIGS. 5-8, the modular welding positioner **10** is shown in a first configuration using the rotary positioner **14**. The rotary positioner **14** is positioned and secured between the guides **24** to permit the rotary positioner **14** to rotate. The rotary positioner **14** includes first and second spaced-apart rings **42** and **44** connected in a stacked configuration by spacers **46** positioned therebetween, a ring gear **48** connected to a bottom **50** of the first ring **42**, and arms **52** pivotally connected to a top **54** of the second ring **44**. As shown, the rotary positioner **14** includes three arms **52**; however, it should be appreciated that any suitable number of arms **52** may be used.

(17) Each of the arms **52** include a jaw **56** positioned at a distal end of the arm **52** to clamp onto a workpiece being welded. As shown, the jaws **56** are in the form of an elongated pin or cylinder with first and second ends **60** and **62** that extend above and below the arms **52**. It should be appreciated that any suitable type of jaw may be used. This permits the jaws **56** to be used to clamp a workpiece “W” between the jaws **56** on an outside of the workpiece or to clamp a workpiece on an inside of the workpiece by moving the arms **52** outwardly and exerting pressure along an inside surface of a wall of the workpiece. As shown in FIGS. 7 and 8, the arms pivot about pivot points **58** to allow the jaws **56** to clamp and hold a workpiece of varying diameters and/or dimensions.

(18) Once a workpiece is clamped by the jaws **56**, the gear **30** engages ring gear **48** and the motor **28** is turned on to rotate the gear **30**. This in turn rotates the rotary positioner **14** and workpiece clamped by the jaws **56** to allow a welder to apply a weld circumferentially around the workpiece. Alternatively, one or more of the guides **24** may be motorized to act as a driving element to drive and/or rotate the rotary positioner **14** instead of using gear **30**.

(19) Referring to FIGS. 9-11, the modular welding positioner **10** is shown in a second configuration using the linear positioner **16**. The linear positioner **16** is positioned and secured between the guides **24** to permit the linear positioner **16** to move linearly therebetween. The linear positioner **16** includes a table **64** and a rack gear **66** connected to a bottom **68** of the table **64**. The table **64** is able to translate linearly between a first direction and a second direction (as indicated by arrows). In use, a welder places a workpiece on the table **64** and applies a linear weld to the workpiece by moving the table.

(20) The modular welding positioner allows a welder to switch between rotary and linear welding by swapping out the positioners **14**, **16**. For example, a welder may want to weld a tank by first welding a sidewall of the tank and then welding end caps to the sidewall. In this example, the welder would first install the linear positioner **16** onto the base **12**. The linear positioner **16** would be positioned between the guides **24** so that the table **64** can move linearly. The gear **30** would then be placed into engagement with rack gear **66**. Alternatively, one or more of the guides **24** may be motorized to act as a driving element to drive and/or move the linear positioner **16** instead of using gear **30**. The welder then places a tubular sheet onto the table **64** so that a linear weld can be made between opposing ends of the sheet. The welder begins the weld and controls the speed and direction of the table **64** so that the tubular sheet moves as the weld is being performed as opposed

to the welder moving. The direction and speed may be controlled by a controller **70**, such as a foot pedal. Additionally, the linear movement may be continuous or stepped.

(21) Once the linear weld is complete, the welder removes the linear positioner **16** from the base and installs the rotary positioner **14**. Gear **30** engages ring gear **48**. The welded tube is placed between the jaws **56** and the arms **52** are rotated until the jaws **56** clamp the welded tube therebetween. It should be appreciated that the arms may be manually rotated or automatically rotated by a motor driving gears, linkages, cables, etc.

(22) The end cap is then placed on the end of the welded tube and the welder initiates welding. The welder rotates the welded tube and cap using the controller **70** which causes the gear **30** to rotate thereby causing the rotary positioner **14** and welded tube to rotate. As stated earlier, the welder can control the speed and rotation direction of the welding positioner **14**. Additionally, the rotation may be continuous or stepped. Once the cap is welded to the welded sheet, the welder disengages the jaws **56** from the welded tube and the welded tube and cap are flipped over to allow a cap to be welded to the opposing end of the welded tube. The welding process is repeated and the a tank is now formed.

(23) The foregoing has described a modular welding positioner. All of the features disclosed in this specification (including any accompanying claims, abstract and drawings), and/or all of the steps of any method or process so disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive.

(24) Each feature disclosed in this specification (including any accompanying claims, abstract and drawings) may be replaced by alternative features serving the same, equivalent or similar purpose, unless expressly stated otherwise. Thus, unless expressly stated otherwise, each feature disclosed is one example only of a generic series of equivalent or similar features.

(25) The invention is not restricted to the details of the foregoing embodiment(s). The invention extends any novel one, or any novel combination, of the features disclosed in this specification (including any accompanying claims, abstract and drawings), or to any novel one, or any novel combination, of the steps of any method or process so disclosed.

Claims

1. A modular welding positioner, comprising: a rotary positioner; a base, the rotary positioner being positioned on the base, the base including: spaced-apart guides positioned thereon, the guides being arranged so as to define both a circular path and a linear path, the rotary positioner being secured between the spaced-apart guides and adapted to rotate therebetween; and a driving element to move the rotary positioner positioned on the base.
2. The modular welding positioner according to claim 1, wherein the driving element is a gear configured to engage a mating gear on the rotary positioner.
3. The modular welding positioner according to claim 2, wherein the base further includes a gate, the gate being pivotally connected to a sidewall of the base to allow the gear to engage and disengage the mating gear of the rotary positioner.
4. The modular welding positioner according to claim 1, wherein the driving element is at least one of the spaced-apart guides.
5. The modular welding positioner according to claim 1, wherein the rotary positioner further includes a plurality of arms, each of the arms including a jaw positioned at a distal end thereof to clamp a workpiece in the rotary positioner.
6. The modular welding positioner according to claim 5, wherein each of the plurality of arms are pivotally connected to the rotary positioner to allow the plurality of arms to move between a clamped position and an unclamped position.
7. A modular welding positioner, comprising: a rotary positioner configured to rotate an object to be welded; a linear positioner configured to move linearly an object to be welded; a base having a

plurality of guides positioned thereon for receiving one of the rotary positioner or linear positioner; a driving element to drive either of the rotary positioner or linear positioner; and wherein the modular welding positioner is configurable between a first configuration where the rotary positioner is positioned between the guides on the base and a second configuration where the linear positioner is positioned between the guides on the base.

8. The modular welding positioner according to claim 7, wherein the base includes: a sidewall; a platform supported by the sidewall, wherein the plurality of guides are spaced along a periphery of the platform; and a gate pivotally connected to the sidewall, wherein the driving element is connected to the gate.

9. The modular welding positioner according to claim 7, wherein in the first configuration, the rotary positioner is rotated between the plurality of guides by the driving element.

10. The modular welding positioner according to claim 7, wherein the rotary positioner includes a plurality of arms pivotally connected to the rotary positioner, each of the arms including a jaw positioned at a distal end of thereof to clamp a workpiece in the rotary positioner.

11. The modular welding positioner according to claim 7, wherein in the second configuration, the linear positioner is moved linearly between the plurality of guides by the driving element.

12. The modular welding positioner according to claim 7, wherein the driving element is a gear configured to engage a mating gear of either the rotary positioner or linear positioner.

13. The modular welding positioner according to claim 12, wherein the base further includes a gate, the gate being pivotally connected to a sidewall of the base to allow the gear to engage and disengage the mating gear.

14. The modular welding positioner according to claim 7, wherein the driving element is at least one of the plurality of guides.

15. A modular welding positioner, comprising: a linear positioner; a base, the linear positioner being positioned on the base, the base including: spaced-apart guides positioned thereon, the guides being arranged so as to define both a circular path and a linear path, the linear positioner being secured between the spaced-apart guides and adapted to move linearly therebetween; and a driving element to move the linear positioner positioned on the base.

16. The modular welding positioner according to claim 15, wherein the linear positioner is moved linearly between the plurality of guides by the driving element.

17. The modular welding positioner according to claim 15, wherein the driving element is a gear configured to engage a mating gear of the linear positioner.

18. The modular welding positioner according to claim 17, wherein the base further includes a gate, the gate being pivotally connected to a sidewall of the base to allow the gear to engage and disengage the mating gear.

19. The modular welding positioner according to claim 15, wherein the driving element is at least one of the plurality of guides.
